

Project Report: Design and Implementation of an IoT-Based High-Precision Environmental Data Acquisition System

1. Introduction

The objective of this project is to design, develop, and deploy an Internet of Things (IoT) based, high-precision environmental monitoring system. While macro-meteorological stations provide broad climate data, there is a growing necessity for hyper-local, real-time micro-climate monitoring in sectors such as precision agriculture, industrial automation, and smart habitats. Traditional Data Acquisition (DAQ) systems are often cost-prohibitive, bulky, and lack seamless wireless integration.

This project addresses these limitations by engineering a compact, low-power telemetry node centered around the ESP32 System-on-Chip (SoC). The system integrates a suite of advanced Microelectromechanical Systems (MEMS) sensors to acquire multi-parameter environmental telemetry—specifically ambient temperature, barometric pressure, ultraviolet (UV) irradiance, and luminous emittance (Lux). By leveraging the ESP32's native IEEE 802.11 (Wi-Fi) capabilities, the system operates as a standalone asynchronous web server, disseminating serialized telemetry data in real-time to any network-connected client via a dynamic, low-latency web interface.

2. Literature Review

A contemporary review of embedded environmental telemetry systems reveals a definitive transition from monolithic, centralized Data Acquisition (DAQ) infrastructures (such as legacy SCADA systems) toward distributed, edge-computing paradigms. The proliferation of Microelectromechanical Systems (MEMS) has enabled the integration of highly sensitive transducer arrays into compact, low-power Application-Specific Integrated Circuits (ASICs).

Within this domain, intra-board communication is overwhelmingly dominated by the Inter-Integrated Circuit (

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) protocol. Originally codified by NXP (formerly Philips),

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remains the de facto standard for short-distance micro-controller peripheral interfacing due to its synchronous, half-duplex, multi-drop architecture utilizing a simple open-drain topology with pull-up resistors.

However, a critical vulnerability persists in modern SoC deployments, particularly concerning continuous-time (analog) signal acquisition. The ESP32 SoC, while highly proficient in RF communications and multi-core processing, is critically bottlenecked by its internal 12-bit Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC). Extensive academic testing demonstrates that the ESP32's native ADC exhibits significant Differential Non-Linearity (DNL) and Integral Non-Linearity (INL), alongside a notably high noise floor, rendering it unsuitable for precision radiometric measurements such as UV irradiance.

State-of-the-art literature mitigates this quantization error by employing external, high-resolution Delta-Sigma (

$$\Delta\Sigma$$

) ADCs. The

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topology provides superior noise shaping and oversampling capabilities compared to standard SAR architectures. By implementing a Texas Instruments ADS1115 16-bit

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ADC at the edge to digitize the analog UV sensor, this project directly applies current literature recommendations to ensure high-fidelity, linear signal

conversion prior to digital transmission over the

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bus.

3. Working Principle

The fundamental working principle of the system relies on the orchestrated transduction of physical and radiometric environmental phenomena into the discrete digital domain, governed by a central deterministic computational node.

The system acts as a multi-modal cyber-physical interface:

- **Thermodynamic and Barometric Transduction (BMP280):** Ambient thermal energy and atmospheric pressure dictate the physical state of the micro-scale piezoresistive elements within the BMP280. Variations in localized atmospheric mass apply mechanical stress to a silicon membrane, altering its electrical resistance. An internal mixed-signal ASIC applies temperature-compensated algorithms to these analog resistance changes, yielding highly accurate, discrete thermal and barometric coefficients.
- **Radiometric Transduction (APDS9930 & Analog UV Sensor):** Luminous emittance is measured via the photoelectric effect. Photons striking the specialized photodiode arrays within the APDS9930 and the analog UV sensor liberate electron-hole pairs, generating a measurable photocurrent proportional to the incident irradiance. The APDS9930 utilizes dual photodiodes to approximate the human eye's photopic response, while the analog UV sensor utilizes a specific semiconductor bandgap to isolate the ultraviolet spectrum (approx. 280-390 nm).
- **Continuous-to-Discrete Domain Conversion (ADS1115):** The continuous analog voltage output from the UV sensor is fed into the differential input of the ADS1115. A modulator converts this continuous-time analog signal into a high-frequency bitstream, which is subsequently decimated by a digital low-pass filter, resulting in a precise 16-bit discrete representation of the UV irradiance.

$$\Delta\Sigma$$

- **Systemic Orchestration:** The ESP32 acts as the central orchestrator, bridging these physical transductions with the cyber domain. Utilizing an asynchronous, non-blocking architecture, the microcontroller synchronously polls these discrete registers via the bus at deterministic intervals, packages the telemetry into standardized JSON payloads, and serves it over a localized TCP/IP socket.

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4. Design Flow

The system architecture is structured as a robust, fault-tolerant state machine, designed to prioritize continuous operation and non-blocking asynchronous event handling. The firmware execution follows a strict top-down topological flow:

1. State 0: Power-On Reset (POR) & Hardware Initialization

- System clock stabilization.
- Initialization of the Universal Asynchronous Receiver-Transmitter (UART) interface at 115200 baud for telemetry debugging.
- Instantiation of the hardware peripheral, defining precise clock stretch limits and assigning SDA to GPIO 33 and SCL to GPIO 32.

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1. State 1: Peripheral Handshake & Vector Verification

- The master transmits target address vectors (0x76 for BMP280, 0x48 for ADS1115, 0x39 for APDS9930) across the bus.
- *Error Handling:* If an ACK (Acknowledge) bit is not received, the system flags a bus fault and continues initialization of available sensors to prevent a total system halt.
- Calibration registers are read and Infinite Impulse Response (IIR) filtering coefficients are injected into the MEMS sensors.

1. State 2: Network Stack Provisioning

- The 802.11 b/g/n RF physical layer is powered on.

- The LwIP (Lightweight IP) stack is configured for Station (STA) mode.
- The system executes an asynchronous blocking loop until a DHCP lease (IP address) is successfully negotiated with the local access point.

1. **State 3: Asynchronous HTTP Server Instantiation**

- URI (Uniform Resource Identifier) endpoints are mapped to execution callbacks:
- GET / Triggers the execution of the INDEX_HTML string payload stored in PROGMEM.

→

- GET /data Triggers the dynamic synthesis of the JSON telemetry payload.

→

- The TCP listener daemon is initiated on Port 80.

1. **State 4: The Super-Loop (Continuous State)**

- **Task A (High Priority):** server.handleClient() continuously monitors the TCP socket for incoming HTTP requests.
- **Task B (Scheduled):** A non-blocking millis() delta-timer tracks elapsed execution time. Upon reaching the 2000ms threshold, a sequential hardware interrupt is simulated: polling BMP280, APDS9930, and ADS1115 over . Raw binary returns are bit-shifted, scaled by their respective sensitivity factors, and stored in global memory buffers ready for the next /data HTTP request.

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5. Circuit Diagram & Connections

A robust hardware design requires precise attention to power stabilization and logic-level definitions. The complete system topology, including necessary passive components, is defined as follows:

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	📎 Files
Primary Power Supply	<u>5V USB IN</u>	ESP32 VIN	Primary power source from USB host.	
Logic Power Bus	<u>3.3V OUT</u>	ESP32 3V3 Pin	The internal LDO regulator steps down 5V to 3.3V. This pin supplies the VCC rail for all peripheral sensors.	
System Ground	<u>GND</u>	Common Ground Rail	All GND pins (ESP32, sensors, external ADCs) must terminate at a common low-impedance ground plane to prevent ground loops.	
Serial Data	<u>SDA</u>	GPIO 33 (D33)	Requires a pull-up resistor to 3.3V. Multiplexed to SDA pins of BMP280, APDS9930, and ADS1115.	

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				<u>tj+6cFQS2OK</u> <u>VJUUVKblf1uVt</u> <u>1cNH2ITWbW</u> <u>dLOdNt3SNPu</u> <u>c25tw03tv2y.</u> <u>SUMdZwvjm/</u> <u>3znn/u73d87v</u> <u>cR3Kefi78Eg3</u> <u>NTVd7vP5vgL</u> <u>HwGmwu7Gx</u> <u>ceViF6/gTDc0</u> <u>NKxOpVKvgY</u> <u>cGBwfrlLoWb</u> <u>Ein0x97PJ4q2</u> <u>osmBZNWVfV</u> <u>2WN3hdDqFr</u> <u>ALxgxB+C9u6</u> <u>kpKSjeh8xRkl</u> <u>BGqpVp2gpg</u> <u>ZmmXGAlpaW</u> <u>EhJ1NTbVRFo</u> <u>eAD8QELYCv</u> <u>u9lcEgvZwA3</u> <u>uAJk5lg0pge</u> <u>nxS7N+aD6/f</u> <u>5mxuwCZ6am</u> <u>pml8MAh6ysv</u> <u>Lx7F11tfXVwr</u> <u>heDz+Ju98DI/</u> <u>aRJpyn8a5Hb</u> <u>wAosALPOBL</u> <u>h8PRRjZfdbvd</u> <u>J6uqq5wuVw</u> <u>rqgur32UgTfB</u> <u>fKg30v6LnAk</u> <u>uoHIJ7mGAPc</u> <u>dcw/INUzU3F</u> <u>LgK1VEsmHm</u> <u>esvSMjlzu12Q</u>

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				<u>K+A4qJ9PDw</u> <u>8ImBgYFPGW</u> <u>QPAaNARErfh</u> <u>v3DcDj8TXd3</u> <u>d1LQ29t7XLQ</u> <u>ESDmZ0CO0j</u> <u>yaTyS60rUB2</u> <u>PRM+RMB9PL</u> <u>ONRF3J2O19f</u> <u>X1j2DQJhULj2</u> <u>LfAYxxlhk8xg</u> <u>T/EaSlRgGlks</u> <u>20WtrgR8ieR</u> <u>NtKRUXFozib</u> <u>GfiJSCQyTNt</u> <u>SIHw3js9AJW</u> <u>QeZtzXSVSCv</u> <u>knETiV2644g</u> <u>kwhL25YOL7+</u> <u>KgAag8ODXo</u> <u>u3A2r+VjG2G</u> <u>xPr+/v5f7Olgf</u> <u>C2+90E5sS+</u> <u>R3U7aaWArxE</u> <u>VwToAeJqglz</u> <u>pY0QS1ABSeY</u> <u>wN9oS2EjrW</u> <u>HgrWThQUj8</u> <u>KhsHrJodLMu</u> <u>HuO3Y3egDL</u> <u>KFdtOckjF9hX</u> <u>NmcQnaf9AW</u> <u>WpLkgShn4Z</u> <u>gkAv09MTAyh</u> <u>TQJhH1V4lqX</u> <u>0GKUMSQCb</u> <u>6QbW6z3SNq</u> <u>KsrOxegnGbb</u> <u>ns7Eolk9otusl</u>

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				<u>aMtwKPLJ+D</u> <u>aE0sSUPCuJ7</u> <u>3s3v/06INf0x</u> <u>sJYTbMS2H5</u> <u>CuUfhe2Dvrtk</u> <u>Auis8LNeTFJe</u> <u>EY3HKWdzZp</u> <u>us1VU6Deqsp</u> <u>YT5a9MkCVpl</u> <u>oNxPf+cCTZq</u> <u>iL1l/xYgWd2E</u> <u>3oTtcfQy9HF0</u> <u>VkjC9XTWAA</u> <u>XfT6z7HL/Y7</u> <u>SCnUzQajeNP</u> <u>AU0sSePJrOd</u> <u>/yIp2zGDLEW</u> <u>beCIQLVLBhu</u> <u>nOCFSUznsJ</u> <u>4sqnnXcuzns/</u> <u>pmkhTYuN6Di</u> <u>YSiQVnJWdk</u> <u>Q4fKyfUrigRL</u> <u>KltmMRQCE2I</u> <u>KOe96LuRF+j</u> <u>NjZHpBG1Div</u> <u>V5vgOO0VRlp</u> <u>/QxMpMmKH</u> <u>6cHyPlsuZ7FI</u> <u>w9Yx0k9nhM</u> <u>sdUZvz6uY4E</u> <u>awiqN00hhslo</u> <u>1zHXCBOc9n/</u> <u>AsWXnxAD75</u> <u>EP8L0rr3io8x</u> <u>LAh9gwnli5Q</u> <u>TOJu3iBbLTJe</u> <u>hlaWmp7XUs</u> <u>AQsF6d1PrHx</u>

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				EyU2ofcrStxX 5qmOZvkzAXj Z19nymr0kOa W4yOxiv0zyK cigYDJ7S20W poaGhwyRDy xi6VS4luwGF 8Ojo6Fb8bjK9 E206aYS0k1L UcjinUUboNB GnfG5QIJDY+I WVjynWOq2B JQ2AbT3/BuJ dxgnzCBsucK JhnhPf5+W7u xO+bnJx80vj VNxMx8+9gZ 95EKY7RjYL3 gEaQjLxDO8p t18WNJh/8dA sXlcAyuV9V1R 2M0sz4hyEZ4 /OfgQ7aA9j3M akPWBJPx2K x/+lbioOd+S0 XhBNYXRRiu1 FeaPIOnka+Tx K873kqWsmjd 4EOSI4BOaXu hEMAwvLI73 9iDGJLA+Tcb ENkgQlfg7aw GawE/zJe+cki 1+Tc0Jae3MR f0uki0heXo8u ZTqvdbURvJT pIpKX16NLmc 4rXUUEnwUA AP//dsjcjQAA

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				AAZJREFUAw BSTDhjbcJU+ gAAAABJRU5 ErkJggg==  YlwSBQkaiRriEhclmpj4AUT R8MEIGAJGo 6j4DUlgRlyGG BdICKABJBhD 0CralbbQNVD aTtN2Ov7+45th3rz3pIPAfoA 2599737nnnH uWu4077Rr6 Gw/2ai32eGW vhsr6/X6vz+c rjKKoqChvxM qWIJTch8lqK Y8mCYFAYC5 6L6GTDlImdB aCGtAFwuAY PuSmaMCFn3 eBg+FwuA+d wy6Xaz39Tz0 eT0PSYFHyl7 wFpYW9vb0e 2pRpeHj4HoQ /wdnzONvoB

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				<u>MYPIZWVTUE</u> <u>2SrXMuZaPGi</u> <u>A61djYeE6dZJ</u> <u>g2bVoWtjYS2</u> <u>BFQ5Ha7ZzU</u> <u>0NPjr6+ufQX</u> <u>8J/YmOwVZ</u> <u>WVmbg8Dom</u> <u>KAWjJhwuN5</u> <u>Ry6BcnQbC7</u> <u>u1tViljVDUOfs</u> <u>IHA0hD7yBtG</u> <u>DiSfE1PT/8Qg</u> <u>VfBmVAotKCu</u> <u>ru4E/XgKOQb</u> <u>b2dn5KBMti5</u> <u>dOtZ+fn3892f</u> <u>Uh34aNXbRf2</u> <u>EDO1MJf3t7e</u> <u>3kMbl6pUwM</u> <u>cdlljjv9ImpY6</u> <u>OjocReAGl3mt</u> <u>qavpHnUTYB</u> <u>qvli7OrcPSDRl</u> <u>VUvrOzsyegm</u> <u>43sfKr0ONVa</u> <u>Hg+W2LuMiV</u> <u>YzXq9OPBgP</u> <u>8F0E6gi2jtaR/</u> <u>BxEDD4PXCD</u> <u>lajxKa0uWYL</u> <u>UkUFiHs7uZ6</u> <u>GdbrRGYOKs</u> <u>9el6EnU0UNe</u> <u>xvwP5+EvBV</u> <u>4rjxXUWrM6K</u> <u>6paUI6X5ljlxk</u> <u>K4CoC9/PqG</u> <u>MHS7AsicU4</u>

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				<u>WwQ241DSv</u> <u>WJnULzJkyf/</u> <u>MTQ09AQHg</u> <u>8VR7L+CTC7</u> <u>2dR7Y2fcw72</u> <u>xktF/30g4Bka</u> <u>u0tPQmDqFIH</u> <u>HaVMFRJGhO</u> <u>FOHVDJk7chy</u> <u>lYrgsfmVrN+F</u> <u>u1tbXnaS+Jjh</u> <u>8/Psi+6ULZFA</u> <u>z27ySQFfBXO</u> <u>9mP269DVOI</u> <u>3ZNPg6aTdw</u> <u>ErbIX8V2NjD8</u> <u>p2nsUH+aKN</u> <u>7PpckalXBslJ</u> <u>8sB6MrUH4a/</u> <u>bRL1bRy+MYh</u> <u>9Y7OPsjy/c3J</u> <u>2vMP4OxEnC</u> <u>K1fEn1ZxltT5D</u> <u>7wR4ikCfZay</u> <u>QREygTWOZd</u> <u>9LuB6lJ8G9U</u> <u>xw6xYluLix9io</u> <u>pkkSveqqSJ2i</u> <u>qPlcWg9gLNV</u> <u>YCe6yexHI2g1</u> <u>gYVx/iN0vmF</u> <u>LbMW/W9Cd</u> <u>CgYY66AVyd</u> <u>Z2OgNApPtdr</u> <u>QWRYHWpY+</u> <u>h1Kru2ublZmb</u> <u>lIXg5DyxCHI2</u> <u>PjLPfh37S2xN</u> <u>L0EsRcY7AN2</u>

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				<u>e0Eu42VEDnl</u> <u>gsHgHuyskAy</u> <u>8l4ZcGv1j9Lc</u> <u>BUZVWkTqJU</u> <u>LAugnyDgX2n</u> <u>T5/+ifaKE07P</u> <u>wsF7MVzX19</u> <u>cX3V98mollm</u> <u>wdnJhCtRKeP</u> <u>lvToBBajtW1l</u> <u>wpvYZvpelFF</u> <u>xRaGSIKuSVX</u> <u>7Nq/X6xczEW</u> <u>6yOQ/B58B0T</u> <u>rqVnHSPRcFE</u> <u>0SUxNSsra4n</u> <u>4VEn3X5ydIL</u> <u>qLkMrE+WY5</u> <u>TN+WmC96v/</u> <u>6FwDLk9Tg5z</u> <u>AlejZ+z4DkSS</u> <u>dBhth6BSSRX</u> <u>VxddM7lZJh4</u> <u>C1RVxN+0ahj</u> <u>dFwffL9EXTm</u> <u>ViGNuGQDhD</u> <u>xUoKxpCKTY</u> <u>69pBCXJ6X4</u> <u>9ydLcRQUXM</u> <u>K9WXRm6r6G</u> <u>rMZqLxAl/K0</u> <u>V4EY6LVfM57</u> <u>QFkF/v/e2zwe</u> <u>ZHcZGQfKLE</u> <u>DSm8boocwd</u> <u>LMhs8/gRZry</u> <u>8nKdio6/bFgR</u> <u>ugoi72TsdUe</u> <u>U7P/Z3a86oE</u>

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				4a4vkFBQXX KXkE9wPBzB efYg2SkEfgTd Kzk76KUgHfci prz0rnUuBh2 X/c39/fzcTfy QkHI3rh3OAw FmNTle3X22 GY3sM4Px2e qF5bwNiP3oG BgUgSuD3aS WlecnPkAyyv MMLNfEcSTD 9GdsGm63Rm ch8KugYkHMj JyakguEIC80 YYgYCce1B9J quighbjGsOG Kqt3sj4dweG ku7UQAdN7m ApFH/X60eBi Py4loB3crzq M0jlzM3uYQ/1 vud4ucNhqS8 7BTuJ2c1mCJ aApTHyMSfR AX4qSqISAILE W+PeLwU+o NtrNgKHwVrK qXzF8monBC 3AiTz6clCN8 WokgyuBm4f heAoldfxkZGQ fgfw/exLd67M 3Azg6+l8S+7 qdjeaQglwE/R myBbEuwHAX ngA+47MC+3 WNYCDO+EW

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				jfr9QT0eCbGp JwlACeZvIVJ HGrATDuly8v b7fH4ymgXQ U7dq3oWckci xjzg9n0nxQP mRjBWwtM/u Ln+zEB0toCl mDhX1FSEsj+ Dibfwns56GR ccjU1NW1qbW RCGhMYc3zo M5aU/vdgk84 +xoPjwY5xws dsuvHKjImqx3 iifwEAAP//UZj fKwAAAAZJR EFUAwDJkgV qk6DafwAAA ABJRU5ErkJg gg≡≡
Serial Clock	<u>SCL</u>	GPIO 32 (D32)	Requires a pull-up resistor to 3.3V. Master clock line generated by ESP32.	

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				<u>c25tw03tv2y</u> <u>SUMdZwvjm/</u> <u>3znn/u73d87v</u> <u>cR3Kefi78Eg3</u> <u>NTVd7vP5vgL</u> <u>HwGmwu7Gx</u> <u>ceViF6/gTDc0</u> <u>NKxOpVKvgY</u> <u>cGBwfrlLoWb</u> <u>Ein0x97PJ4q2</u> <u>osmBZNWVfV</u> <u>2WN3hdDqFr</u> <u>ALxgxB+C9u6</u> <u>kpKSjeh8xRkl</u> <u>BGqpVp2gpq</u> <u>ZmmXGAIPA W</u> <u>EhJ1NTbVRFo</u> <u>eAD8QELyCv</u> <u>u9lcEgyZwA3</u> <u>uAJk5lg0pqe</u> <u>nxS7N+aD6/f</u> <u>5mxuwCZ6am</u> <u>pml8MAh6ysv</u> <u>Lx7F11tfXVwr</u> <u>heDz+Ju98DI/</u> <u>aRJpyn8a5Hb</u> <u>wAosALPOBL</u> <u>h8PRRjZfdbvd</u> <u>J6uqqt5wuVw</u> <u>rqqr32UgTfB</u> <u>fKg30v6LnAk</u> <u>uoHIJ7mGAPc</u> <u>dcw/INUzU3F</u> <u>LgK1VEsmHm</u> <u>esvSMjlzu12Q</u> <u>K+A4qJ9PDw</u> <u>8ImBgYFPGW</u> <u>QPAaNARErfh</u> <u>v3DcDj8TXd3</u>

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	📎 Files
				d1LQ29t7XLQ ESDmZ0CO0j yaTyS60rUB2 PRM+RMB9PL ONRF3J2O19f X1j2DQJhULj2 LfAYxxlhk8xg T/EaSlRgGlks 20WtrgR8ieR NtKRUXFozib GfiJSCQyTNt SIHw3js9AJW QeZtzXSVSCv knETiV2644g kwhL25Y0L7+ KgAag8ODXo u3A2r+VjG2G xPr+/v5f7Olgf C2+90E5sS+ R3U7aaWArxE VwToAeJqglz pY0QS1ABSeY wN9oS2EjrW HgrWThQUj8 KhsHrJodLMu HuO3Y3egDL KFdtOckjF9hX NmcQnaf9AW WpLkgShn4Z gkAv09MTAyh TQJhH1V4lqX 0GKUMSQCb 6QbW6z3SNq KsrOxeqnGbb ns7Eolk9otusl aMtwKPLJ+D aE0sSUPCuJ7 3s3v/06INf0x sJYTbMS2H5

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				CuUfhe2Dvrtk Auis8LNeTFJe EY3HKWdzZp us1VU6Deqsp YT5a9MkCVpl oNxPf+cCTZq iL1l/xYgWd2E 3oTtcfQy9HF0 VkjC9XTWAA XfT6z7HL/Y7 SCnUzQajeNP AU0sSePJrOd /yIp2zGDLEW beCIQLVLBhu nOCFSUznsJ 4sqnnXcuzns/ pmkhTYuN6Di YSiQVnJWdk Q4fKyfUrgRL KltmMRQCE2l KOe96LuRF+j NjZHpBG1Div V5vgOO0VRlp /QxMpMmKH 6cHyPlsuZ7FI w9Yx0k9nhM sdUZvz6uY4E awiqN00hhslo 1zHXCBOc9n/ AsWXnxAD75 EP8L0rr3io8x LAh9gwnli5Q TOJu3iBbLTJe hlaWmp7XUs AQsF6d1PrHx EyU2ofcrStxX 5qmOZvkzAXj Z19nymr0kOa W4yOciv0zyK

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				cigYDJ7S20W poaGhwyRDy xi6VS4luwGF 8Ojo6Fb8bjK9 E206aYS0k1L UcjnUUboNB GnfG5QIJDY+I WVjynWOq2B JQ2AbT3/BuJ dxgnzCBsuck JhnhPf5+W7u xO+bnJx80vj VNxMx8+9gZ 95EKY7RjYL3 gEaQjLxD08p t18WNJh/8dA sXlcAyuV9V1R 2M0sz4hyEZ4 /OfgQ7aA9j3M akPWBJPx2K x/+lbioOd+S0 XhBNYXRRiu1 FeaPIOnka+Tx K873kqWsmjd 4EOSI4BOaXu hEMAwvLI73 9iDGJLA+Tcb ENkgQlfg7aw GawE/zJe+cki 1+Tc0Jae3MR f0uki0heXo8u ZTqvdbURvJT plpKX16NLmc 4rXUUEnwUA AP//dsjcjQAA AAZJREFUAW BSTDhjbcJU+ gAAAABJRU5 ErkJggg== da

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	📎 Files
				ta:image/png;base64,iVBORw0KGgoAAAANSUhEUgAAADsAAAAeCAYAAACSRGY2AAAGYUIEQVR4AeyXaWxUVRTHO9NpnbYiIKaUWtqZaYOVUBVT0WANYlwSBQkaiRriEhclmpj4AUTR8MEIGAJGo6j4DUlgRlyGBdICKABJBhD0CralbbQNVDaTtN2Ov7+45th3rz3plPAfoA2599737nnnHuWu4077Rr6Gw/2ai32eGWvhsr6/X6vz+crrjKKoqChvxMqWlJTch8lqKY8mCYFAYC56L6GTDIImdBaCGtAFwuAYPuSmaMCFn3eBg+FwuA+dwy6Xaz39Tz0eT0PSYFHyl7wFpYW9vb0e2pRpeHj4HoQ/wdnzONvoBMYPIZWVTUE2SrXMuZaPGiA61djYeE6dZJg2bVoWtjYS2

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				<u>BFQ5Ha7ZzU</u> <u>0NPjr6+ufQX</u> <u>8J/YmOwVZ</u> <u>WVmbg8Dom</u> <u>KAWjJhwuN5</u> <u>Ry6BcnQbC7</u> <u>u1tViljjVDUOfs</u> <u>IHA0hD7yBtG</u> <u>DiSfE1PT/8Qg</u> <u>VfBmVAotKCu</u> <u>ru4E/XgKOQb</u> <u>b2dn5KBMti5</u> <u>dOtZ+fn3892f</u> <u>Uh34aNXbRf2</u> <u>EDO1MJf3t7e</u> <u>3kMbl6pUwM</u> <u>cdlljjv9ImpY6</u> <u>OjocReAGl3mt</u> <u>qavpHnUTYB</u> <u>qvli7OrcPSDRI</u> <u>VUvrOzsyegm</u> <u>43sfKr0ONVa</u> <u>Hg+W2LuMiV</u> <u>YzXq9OPBgP</u> <u>8F0E6gi2jtaR/</u> <u>BxEDD4PXCD</u> <u>lajxKa0uWYL</u> <u>UkUFiHs7uZ6</u> <u>GdbrRGYOKs</u> <u>9el6EnU0UNe</u> <u>xvwP5+EvBV</u> <u>4rjxXUWrM6K</u> <u>6paUI6X5ljlxx</u> <u>K4CoC9/PqG</u> <u>MHS7AsicU4</u> <u>WwQ241DSv</u> <u>WJnULzJkyf/</u> <u>MTQ09AQHg</u> <u>8VR7L+CTC7</u>

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				2dR7Y2fcw72 xktF/30g4Bka u0tPQmDqFIH HaVMFRJGhO FOHVDJk7chy lYrgsfmVrN+F u1tbXnaS+Jjh 8/Psi+6ULZFA z27ySQFfBXO 9mP269DVOI 3ZNPg6aTdw ErbIX8V2NjD8 p2nsUH+aKN 7PpckalXBslJ 8sB6MrUH4a/ bRL1bRy+MYh 9Y7OPsjy/c3J 2vMP4OxEnC K1fEn1ZxltT5D 7wR4ikCfZay QREygTWOZd 9LuB6IJ8G9U xw6xYluLix9io pkkSveqqSJ2i qPlcWg9gLNV YCe6yexHI2g1 gYVx/iN0vmF LbMW/W9Cd CgYY66AVyd Z2OgNAPtdr QWRYHWpY+ h1Kru2ublZmb lIXg5DyxCHI2 PjLPfh37S2xN L0EsRcY7AN2 e0Eu42VEDnl gsHgHuyskAy 8l4ZcGv1j9Lc BUZVWkTqJU

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				LAugnyDgX2n T5/+ifaKE07P wsF7MVzX19 cX3V98mollm wdnJhCtRKeP lvToBBejtbW1l wpvYZvpelFF xRaGSIKuSVX 7Nq/X6xczEW 6yOQ/B58B0T rqVnHSPRcFE 0SUxNSsra4n 4VEn3X5ydIL qLkMrE+WY5 TN+WmC96v/ 6FwDLk9Tg5z AlejZ+z4DkSS dBhth6BSSRX VxddM7lZJh4 C1RVxN+0ahj dFwffL9EXTm ViGNuGQDhD xUoKxpCKTY 69pBCXJ6X4 9ydLcRQUXM K9WXRm6r6G rMZqLxAl/K0 V4EY6LVfM5Z QFkF/v/e2zwe ZHcZGQfKLE DSm8boocwd LMhs8/gRZry 8nKdio6/bFgR ugoi72TsdUe U7P/Z3a86oE 4a4vkFBQXX KXkE9wPBzB efYg2SkEfgTd Kzk76KUgHfci

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				prz0rnUuBh2 X/c39/fzcTfy QkHI3rh3OAw FmNTIe3X22 GY3sM4Px2e qF5bwNiP3oG BgUgSuD3aS WlecnPkAyyv MMLNfEcSTD 9GdsGm63Rm ch8KugYkHMj JyakguEIC80 YYgYCce1B9J quighbjGsOG Kqt3sj4dweG ku7UQAdN7m ApFH/X60eBi Py4IoB3crzq M0jlzM3uYQ/1 vud4ucNhqS8 7BTuJ2c1mCJ aApTHyMSfR AX4qSqISAILE W+PeLwU+o NtrNgKHwVrK qXzF8monBC 3AiTz6clCN8 WokgyuBm4f heAoldfxkZGQ fgfw/exLd67M 3Azg6+l8S+7 qdjeaQglwE/R myBbEuwHAX ngA+47MC+3 WNYCDO+EW jfr9QT0eCbGp JwIACeZvIVJ HGraTDuly8v b7fH4ymgXQ

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	📎 Files
				U7dq3oWckci xjzg9n0nxQP mRjBWwtM/u Ln+zEBOtoCl mDhX1FSEsj+ Dibfwns56GR ccjU1NW1qbW RCGhMYc3zo M5aU/vdggk84 +xoPjwY5xws dsuvHKjImgx3 iifwEAAP//UZj fKwAAAAZJR EFUAwDJkgV qk6DafwAAA ABJRU5ErkJg gg==
Analog UV Sensor Power	<u>VCC</u>	3.3V Rail	Powers the internal operational amplifier of the UV sensor module.	
Analog UV Sensor Signal	<u>OUT</u>	ADS1115 A0	Analog voltage routed directly into the positive differential input of the 16-bit ADS1115.	
Decoupling (Optional but recommended)	<u>/ GND</u>	Ceramic Caps	Placed physically close to the VCC/GND pins of each sensor IC to filter high-	

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
			frequency switching noise.	4AeyXW2ibZRjHkyZNa622Ww/BJkuTloJbvVDiisoQpgwmgl6lCN4ok4mgA6+cU5grOLeJlgxP83AjDlfUebELFWUTL4brLPWwOoNNmkNbbVo3nZ21tW38/T/7pfnyfd2BJoPByvPP877P6X3e5z19rXBdJn9XEi31Ql2paFkq2trauikUCqXhX4K3hHA43AOPg9cZ1Avy.1NbWVof8kHwKgeyLQCDQkDcsYcNYerfbPVBRUfEUcafBZiGXyzXBt8/Nzb0BnwV5muEPn/3gDFgF6vD/AL5zZGTkz7xhCRtGoslk8hg4QNxxgUEk+koqlDo/PDz8oyEo+EE2JXsS+wjxESYTob8VfEXfMin6JSEjUTMSy.WVp/w1Ea/SzFFjyTnQPkuw2Ej9Fu6xkSZTKjDHaaeAigYC4E6LRaCX655hYD1Xsdblp

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				<u>tcySqM/n+5c</u> <u>BtE9hLh0Ktxr</u> <u>FyGazdyHrmJ</u> <u>2d3QfPgbKTJ</u> <u>dHJycmzjJgB</u> <u>okBTU9PVahS</u> <u>ivb29mYOzg4</u> <u>ru5uAMF+rK2</u> <u>bYkOj4+rKRH</u> <u>FgZcVVtbW5y</u> <u>omypuQT/Y0</u> <u>NBwEH7JyJlo</u> <u>o2oZf4eLqriF</u> <u>KtUwwd3aRS</u> <u>UfYG929/X1a</u> <u>ZuYqrLz4kRdJ</u> <u>GFWdIXH4/G</u> <u>bGQSDwatoP</u> <u>4N+XzqdHqBt</u> <u>ISZRz4V/nRO</u> <u>0XTD2ABvpY</u> <u>EYiEb+Tn2Rsv</u> <u>1o52RJF+BM</u> <u>Q1VC9ZjUE9u</u> <u>V9JFIH8u+o7</u> <u>4CNyPSKaaK</u> <u>HaHcv4Hm2y</u> <u>1EGjTGZG5FZ</u> <u>aGJionF+fl6P</u> <u>zVEUSfASMH</u> <u>wZ/82amppT4</u> <u>XB4j1Oiup7M</u> <u>S3sITi4Mwzht</u> <u>I9GdiUTC8eX</u> <u>hmjqA/jXZgxd</u> <u>4LB5dwCONj</u> <u>Y2rkR1G//5Cd</u> <u>en+T9j8Suxn6</u> <u>alAvdXV1Y8h</u>

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	📎 Files
				M3yJeS8FWo ffZqdEp3CaA doGQbiXGW/ F+GuW/Aj9c1 EUpSZ6Ep4n7 Wdivl2gjepug Fslma5CPSD9 sVjsr0LI0NDQ cfqf2hJldqMo JoCL2azhBVq P7Hbae5CZla Zply92a5Emm NRvcAvhr6tsH OF6YCF0EQR 6YI7Bi4IwuR9 siSL9B0s9pap oJ0FeRraLZd D+QeVMHAh V5Sa031N5VZ XmIhFH+30Fk 7HcJLJA1gWf wuYE3EboO2 yJTk9Pa9nNf RglyRTL9rHN u0jA89uBKAQ ctwcxWtBdQz xVImae9AI5G 70YMdJwC/n 9ft3IYVuiY2Nj Z5mBTq4c/uB nOx8dU/BzEt XQHpskhZ+X MFwnOQlblel paUe+Wr8TrA SGo/ulIVVVe mK7LQIKhOC 6QRq6XdxAr+ T7Dxw43MLN kkORgpulX1ol

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				7gTDPa98Q0 8T16vt5VOmll KrgEh7ilH17b wOybK7A5T1 W6CXNBHB1 VpwF4BT46Qj tqqwiRuZei1x N0dj8eN/U/fIB K5mYYPG9sjo kcG+cPE/tYx0 Uwmc5zDs6P gziTW0sSEjK pgoUvbcjNwy. KTbi+5dXpke eCGZK5Hmlf mIUeHbzSS2k KSe7SccE8Xo oohg1+Pgo2K F36YerrYNVO RzdJ9w6T+u+ 5R2ntDX43sD gn7uS/Mbw6 X/u3jJ9qLbBO 5A17usRHmx NhJQp/g9BtM 9epDB0wLy0 wzyJHil1Xm6 MEkqW4v+Q3 Rx/LT09+CTA PoHM0t1+5IO prKysgtf44ws K1GCfMZhcWl PcINmTm5loH 0tuBsbnXLLle FzchLd/WAlkJ 9PPgJ9M8aL g4ODZ5ilQctK 1lhwiX4um0T/ AwAA//+FC5E

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				EAAAABkIEQV QDAF45Mlun wxHtAAAAAEI FTkSuQmCC 

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				ryzHdDIYARa AUcJcpqB8A qyAh9bx48cP kzdiSGOCpUz kObAHnLt8+fl +Gt5uWqT51N TUPET1DvBFJ BKpj0ajjymlvB P7HbW1tYvl+ 8TGxufgCgqd nZ27T5w4sZ Kxveu6Ya7PS zcSmgvE3Uy. 74+DvixcvXiI1 YkgzOT4424a ThWTXgLRSV VU1rbi4+CUa/ VpUVLSmo6P DOE2m69Gfo dN1jY2NWjGK oZCNjTHM/6c Flw74nbH/Rh ooEBdIDt+Bs+ SH3EaGNCIge IMsFtvY1dW1F /IMAW6joJTO Wmmn3XiIFex PbUMnJynvA/ Pi8bjakA2FbG yMYX4/Ycbnj unAwMBAVyb 3bKTDqW0M aamKbPMQYI 4eA7iATRx4o kVAfxYFiXMv qREbG2OYx0 91dXUF7m4D kva+vr7zyiQR JqS00ub6ZNk

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				<u>kjuP4dqMVaR</u> <u>UVFbocdN6N</u> <u>0wyfZurDNjb</u> <u>YGWESM5nM</u> <u>EnAXCvdmddj</u> <u>6N6JbQkxe3</u> <u>NTUVENdRm</u> <u>HhGmhUBULs</u> <u>oB+VuqCfCgj</u> <u>aQBvvpqRuJ/</u> <u>gFeGJFGjfJO</u> <u>DyYWMUR/Yt</u> <u>8OolwA5fa2M</u> <u>gppCwidm5n</u> <u>Mg+CXVxUG</u> <u>0UQRO0aHh7</u> <u>ejO5+xrCaMB</u> <u>Cj7llsHJBOWq</u> <u>kMg1PYHiT1B</u> <u>LJ0KuKDg4N</u> <u>nXCWnZguh6</u> <u>8pJo7NSnE4</u> <u>HIQbco/Q/BH</u> <u>5rysrKJtnYsPl</u> <u>TsHuaHfEUnl</u> <u>8E3WAVBO1G</u> <u>9wqTWQh0G</u> <u>y5Ffw08w3PJ</u> <u>jlt8kKTGM4W</u> <u>VVhZICWQvB</u> <u>Z/S18cYHRhx</u> <u>ZFH5xWqn+V</u> <u>1kLIWZYK79G</u> <u>BtWXiFgCpM5</u> <u>SF5kXEdvFJ0</u> <u>XuLS+Jm8Ex</u> <u>SCZf4CxIw0U</u> <u>FiE1nt1Ao1fBG</u> <u>+A18AjQkfX8q</u>

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				<u>hyEXCcT5ON/00FUNc5/gKA/IP4W8pPZ2V+x07aT94Q6tdOD9fwl/ryKE Rn8zUJVC0TO4/itFditsl+EPkYbX9BHN0qsSGNcepKk3jqjHKcoOrnWz9nYEE8+Y0Jrk77uSKZt3d3d2lnJYihEzJtLYQLY39vb+yfpWDLm+4zFKGbHHmGsGZ8gtqTpodenkbEyOkLKjoU4bRKQlrON65C4o2M5n3ICdABPuBBKmbB+mUj3OR/f84eyEbWDFPcn06j3GWMsx8+eTPFMzqxISzo+KgdZIHr69OkLNjaubyar+NNIOcZR9N1k3KAz0d8JdCF9TxxoHOLpVMwGkv3J8ShvwK7emrKrjW6sjxVpSWdtSlmdiFZReRd6k6E3byEmnBpYbWzkVsdPD85D4XDY9+uDH aKffXpXfcukuZgFG0Abt6iCv</u>

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				<u>mwNaJcaz1L</u> <u>HZOpz+VwJa</u> <u>VpVrW4jxJg3</u> <u>m9txSUmJJq</u> <u>EdcJR40+7qS</u> <u>W1sMAuZeMZ</u> <u>C7GZH6JhLF</u> <u>9LikFkMFNi3k</u> <u>sYZywOkB/lp</u> <u>5721KEvGjGe</u> <u>qzAUeaeyWq</u> <u>axQJStUR8fq</u> <u>QH4aJk6cOC</u> <u>+pr9D/2KQU</u> <u>WNVO0jdBM0</u> <u>dkGakDJA6rq</u> <u>neTbqn3jx075</u> <u>gVWG5vm5ub</u> <u>JOL0VKFaLg</u> <u>AqTwYYcw9z</u> <u>qOf4Pon2Q5D</u> <u>gUV2msYNZ1</u> <u>D+MTnKG8Y1</u> <u>DV8kjWc8Y9x</u> <u>eG6rKCRxqt3</u> <u>wa9dBolfRRIN</u> <u>PFvyEh/nDgw</u> <u>h7wnkUjkHQa</u> <u>zCaxnEFuAHo</u> <u>lbVKbRWtWT</u> <u>+kQ66rO2GR</u> <u>oaKsOBxnGYx</u> <u>fHFM8VK6t4D</u> <u>uvk+YSe24/tl</u> <u>FucndCEW/C</u> <u>3SXqAf3Ga3k</u> <u>r+JOf5M2kuc</u> <u>a5sxY4YuGYr</u> <u>Zi0daNBpdRm</u>

≡ Component / Subsystem	Aa Terminal / Pin	≡ ESP32 GPIO / External Connection	≡ Critical Design Notes & Passive Requirements	🔗 Files
				dOGkziTeO7u fRvIOxWE2fq 6VL/1dA/Hncx mOn4Wad69 D6RLhcb/JzC 31yeAq09PT2 +eCbH1G+jLk L+bghrovwleS Pkl4N0c5rNE 0X/WDDts/14 pGVrENROR5 DBvQ5Wgg+C JjfSLgebhPyx y8d8F6qOfk+ yGF68G9lfPst 5IS2fA7oWfF3 dpF2IDBZIs1iY AmkF0iwYsD Ap7LQCaRYM WJgUdlqBNA sGLEz+BQAA/ /+eUhFAAAA ABkIEQVQDA GocO3J1fNK1 AAAAAEIFTkS uQmCC

6. Working of the Model (Step-by-Step Operation)

The operational sequence of the fully assembled system executes in a rigorous, chronological manner:

Step 1: Power Transients & Bootloader Execution

Upon connection to a 5V power source, the ESP32's internal Low-Dropout (LDO) regulator stabilizes the 3.3V power bus. The ESP32's internal bootloader executes from ROM, loading the user firmware from the external flash memory. The

$$100nF$$

decoupling capacitors across the sensor power rails absorb initial inrush currents, ensuring stable logic levels.

Step 2: Bus Arbitration and Sensor Configuration

The ESP32 assumes control as the

$$I^2C$$

master. It pulls the SCL line high and issues a start condition. Sequentially, it pings the BMP280, ADS1115, and APDS9930.

- It commands the BMP280 to load its factory calibration constants (required for pressure compensation).
- It commands the APDS9930 to disable proximity sensing and activate only the Ambient Light Sensor (ALS) engine.
- It configures the ADS1115's Programmable Gain Amplifier (PGA) to map the UV sensor's maximum expected voltage to the full 16-bit scale.

Step 3: Network Association

The ESP32 broadcasts a probe request to the configured Wi-Fi SSID ("Kevin's S24 FE"). Upon successful WPA2 handshake, the router assigns the ESP32 a local IP address, establishing the network route.

Step 4: The Transduction and Polling Loop (Every 2 seconds)

- **BMP280 Polling:** The ESP32 requests 6 bytes of data from the BMP280 (temperature and pressure MSB, LSB, and XLSB). The ESP32 applies the pre-loaded calibration algorithms to these binary values, yielding standard Celsius and Hectopascal floating-point variables.
- **APDS9930 Polling:** The ESP32 requests the CH0 and CH1 ADC channel data from the APDS9930. Using empirical formulas provided in the Avago datasheet, it calculates the Lux value.
- **ADS1115 & UV Sensor Polling:** The ESP32 triggers a single-ended read on channel A0 of the ADS1115. The ADS1115 samples the analog voltage output from the UV sensor, returning a 16-bit signed integer. The ESP32 computes the real voltage (V) and scales it to the standard UV Index scale.

$$Volts = raw \times multiplier$$

Step 5: Asynchronous Client Servicing

A user enters the ESP32's IP address into a mobile browser. The browser sends an HTTP GET / request. The ESP32 intercepts this TCP packet and responds by transmitting the HTML/CSS layout of the "Cyber-Terminal" dashboard.

Step 6: Dynamic DOM Manipulation

The webpage loads on the user's screen. An embedded JavaScript function immediately fires an asynchronous fetch('/data') request. The ESP32 receives this, serializes the latest polled variables from Step 4 into a structured JSON string {"btemp":34.5, "pres":1001.7, "lux": 250, "uv": 0.5}, and transmits it back. The client's JavaScript parses this JSON and injects the specific values into the corresponding HTML ID tags, updating the screen instantly without a visual page refresh. This cycle repeats indefinitely.

7. Results

The prototype was successfully synthesized, deployed, and subjected to operational testing, yielding robust and continuous environmental telemetry.

- **Thermal and Barometric Stability:** The integration of the BMP280 proved highly reliable. Empirical observations during testing demonstrated stable readings (e.g., an operational ambient temperature of 34.5°C and an atmospheric pressure of 1001.7 hPa) with negligible jitter, confirming excellent signal integrity on the communication bus.

$$I^2C$$

- **ADC Precision:** By offloading the UV analog conversion to the TI ADS1115, the system successfully mitigated the voltage non-linearities endemic to the ESP32. This resulted in highly accurate, stable voltage outputs that were successfully scaled to a standardized UV Index.
- **Network Performance:** The asynchronous server architecture demonstrated exceptional resilience. The system successfully maintained a strict 2-second data refresh rate (2000ms polling interval) under continuous client load. The firmware operated without experiencing heap memory fragmentation or Watchdog Timer (WDT) resets, confirming the viability of the system for long-term, unattended deployment.

8. Applications

The implemented system possesses broad applicability across multiple domains:

- **Precision Agriculture & Smart Greenhouses:** Facilitates real-time monitoring of micro-climates, enabling automated control of irrigation and shading systems to maximize crop yield and prevent thermal stress.
- **Industrial HVAC Automation:** Provides localized, high-resolution thermal and barometric data to Building Management Systems (BMS) to optimize heating, ventilation, and air conditioning efficiency.
- **Occupational Health and Safety:** Enables the continuous monitoring of hazardous UV irradiance levels in outdoor industrial, agricultural, or construction environments to ensure worker safety.
- **Meteorological Edge Computing:** Serves as a foundational node that can be deployed en masse to create a high-resolution, hyper-local weather mapping network.

9. Future Scope and Conclusion

Conclusion: This project successfully achieved its objective of designing and implementing a high-performance, IoT-enabled environmental monitoring system. By leveraging the computational power of the ESP32 SoC and adhering to rigorous embedded system design principles—such as utilizing an external 16-bit Delta-Sigma ADC for precision analog measurements and employing an asynchronous, non-blocking web server architecture—the resulting DAQ system is highly accurate, highly responsive, and robust enough for continuous operation.

Future Scope:

The modular nature of the

$$I^2C$$

bus and the dual-core architecture of the ESP32 present significant avenues for future expansion. Future iterations of this project will aim to integrate industrial-grade communication protocols, specifically Modbus RTU over RS485, to interface with heavy-duty sensors designed for harsh environments (such as

the XY-MD01/SHT20). Furthermore, the application layer can be upgraded from a localized HTTP server to an MQTT (Message Queuing Telemetry Transport) client. This enhancement would allow the system to publish telemetry directly to cloud infrastructure (such as AWS IoT or Google Cloud Platform) for long-term time-series data logging, global remote access, and predictive climate modeling utilizing machine learning algorithms.

10. References

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